

FINAL EXAMINATION IN CALCULUS II - SEMESTER: 2023.2

Elitech programs Total time: 90 minutes

Attention: No document is allowed.

Question 1 (1pt): Find the curvature of $x = t + \sin 2t$, $y = 2 - \cos t$ at the point when $t = \pi$.

Question 2 (1pt): Evaluate the double integral $\iint_D 2x dx dy$ where D is defined by $-1 \leq y \leq 1$, $y^2 \leq x \leq 1$.

Question 3 (1pt): Evaluate the surface integral $\iint_S (z^2 + 1) dS$, where S is the surface $z = \sqrt{x^2 + y^2}$, $1 \leq z \leq 2$.

Question 4 (1pt): Calculate the flux of the vector field

$$\vec{F} = (3y^2 - z^2)\vec{i} + (x^2y + z^3)\vec{j} + (x^3 + y^2z)\vec{k}$$

across the outward oriented surface of the solid bounded by the cone $z = 1 - \sqrt{x^2 + y^2}$ and the xy -plane.

Question 5 (1pt): Show that $\vec{F} = (8xe^{2z} + 5y^2)\vec{i} + (10xy - 3y^2e^{2z})\vec{j} + e^{2z}(8x^2 - 2y^3)\vec{k}$ is a conservative vector field. Find a function f such that $\vec{F} = \nabla f$.

Question 6 (1pt): Evaluate $\iiint_V \sqrt{4x - x^2 - y^2 - z^2} dx dy dz$, where V is defined by $x^2 + y^2 + z^2 \leq 4x$.

Question 7 (1pt): Evaluate the following integrals:

a) $\int_0^{+\infty} \frac{x^2 \sqrt{x}}{(x^3 + 1)^2} dx.$

b) $\int_0^{+\infty} \frac{e^{-x} - 3^{-x}}{x} dx.$

Question 8 (1pt): Evaluate $\int_C (6x\sqrt{x^2 + 1} - 2xy^2) dx + (y^3 - 3x^2y^2) dy$, where C is the curve $y = \sqrt{1 - x^3}$ from $A(1; 0)$ to $B(0; 1)$.

Question 9: (1pt). Find the area of the region enclosed by $9x^2 - 4xy + y^2 = 4$.

Wishing you success in your exam!

LỜI GIẢI CHI TIẾT MÔN GIẢI TÍCH II NHÓM NGÀNH CTTT

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Question 1. Find the curvature of $x = t + \sin 2t$, $y = 2 - \cos t$ at the point when $t = \pi$.

[Answer]

+) We have:

$$x(t) = t + \sin(2t), \quad y(t) = 2 - \cos(t)$$

To find the curvature κ of a parametric curve, we use the formula:

$$\kappa = \frac{|x'y'' - y'x''|}{((x')^2 + (y')^2)^{3/2}}$$

+) Compute the derivatives:

$$x'(t) = 1 + 2\cos(2t),$$

$$x''(t) = -4\sin(2t),$$

$$y'(t) = \sin(t),$$

$$y''(t) = \cos(t)$$

At $t = \pi$:

$$x'(\pi) = 1 + 2\cos(2\pi) = 1 + 2(1) = 3,$$

$$x''(\pi) = -4\sin(2\pi) = -4(0) = 0,$$

$$y'(\pi) = \sin(\pi) = 0,$$

$$y''(\pi) = \cos(\pi) = -1$$

+) Now plug into the curvature formula:

$$\kappa = \frac{|x'y'' - y'x''|}{((x')^2 + (y')^2)^{3/2}} = \frac{|3 \cdot (-1) - 0 \cdot 0|}{(3^2 + 0^2)^{3/2}} = \frac{|-3|}{27} = \frac{1}{9}$$

+) Conclusion: $\frac{1}{9}$

Question 2. Evaluate the double integral $\iint_D 2x \, dx \, dy$ where D is defined by $-1 \leq y \leq 1$, $y^2 \leq x \leq 1$.

[Answer]

We have:

$$\begin{aligned}\iint_D 2x \, dx \, dy &= \int_{-1}^1 \left(\int_{y^2}^1 2x \, dx \right) dy = \int_{-1}^1 [x^2]_{x=y^2}^{x=1} dy = \int_{-1}^1 (1 - y^4) dy = \left[y - \frac{y^5}{5} \right]_{-1}^1 \\ &= \left(1 - \frac{1}{5} \right) - \left(-1 + \frac{1}{5} \right) = \frac{4}{5} + \frac{4}{5} = \frac{8}{5}.\end{aligned}$$

Question 3. Evaluate the surface integral $\iint_S (z^2 + 1) \, dS$, where S is the surface $z = \sqrt{x^2 + y^2}$, $1 \leq z \leq 2$.

Consider the surface integral

$$I = \iint_S (x^2 + 1) \, dS \text{ over the surface } S = \{(x, y, z) \mid z = \sqrt{x^2 + y^2}, z \in [1, 2]\}$$

Then the projection of S onto the Oxy plane corresponds to the region

$$D = \{(x, y) \mid 1 \leq x^2 + y^2 \leq 4\}.$$

Transforming to a double integral:

$$I = \iint_D (x^2 + y^2 + 1) \sqrt{\frac{x^2}{x^2 + y^2} + \frac{y^2}{x^2 + y^2} + 1} \, dx \, dy = \iint_D (x^2 + y^2 + 1) \sqrt{2} \, dx \, dy$$

Convert to polar coordinates:

$$\begin{cases} x = r \cos \varphi \\ y = r \sin \varphi \end{cases} \Rightarrow I = \int_0^{2\pi} \int_1^2 (r^2 + 1) \cdot r \cdot \sqrt{2} \, dr \, d\varphi = 2\pi\sqrt{2} \int_1^2 (r^3 + r) \, dr = 2\pi\sqrt{2} \left[\frac{r^4}{4} + \frac{r^2}{2} \right]_1^2$$

$$\begin{aligned}&= 2\pi\sqrt{2} \left(\left(\frac{16}{4} + \frac{4}{2} \right) - \left(\frac{1}{4} + \frac{1}{2} \right) \right) \\ &= 2\pi\sqrt{2} \left(4 + 2 - \frac{3}{4} \right) = \frac{21\pi\sqrt{2}}{2}\end{aligned}$$

Question 4. Calculate the flux of the vector field

$$\vec{F} = (3y^2 - z^2) \vec{i} + (x^2y + z^3) \vec{j} + (x^3 + y^2z) \vec{k}$$

across the outward-oriented surface of the solid bounded by the cone $z = 1 - \sqrt{x^2 + y^2}$ and the xy -plane.

[Answer]

Consider the vector field

$$\vec{F} = (3y^2 - z^2)\vec{i} + (x^2y + z^3)\vec{j} + (x^3 + y^2z)\vec{k}.$$

Apply the Ostrogradsky (Divergence) theorem:

$$\Phi = \iint_S \vec{F} \cdot \vec{n} dS = \iiint_V \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \right) dx dy dz = \iiint_V (0 + x^2 + y^2) dx dy dz$$

with

$$P = 3y^2 - z^2, \quad Q = x^2y + z^3, \quad R = x^3 + y^2z,$$

and S is the boundary of the region V :

$$0 \leq z \leq 1 - \sqrt{x^2 + y^2}$$

Convert to cylindrical coordinates:

$$x = r \cos \varphi, \quad y = r \sin \varphi, \quad z = z$$

$$\begin{aligned} \Phi &= \int_0^{2\pi} \int_0^1 \int_0^{1-r} r^3 dz dr d\varphi = \int_0^{2\pi} \int_0^1 r^3(1-r) dr d\varphi \\ &= \int_0^{2\pi} \left(\int_0^1 (r^3 - r^4) dr \right) d\varphi = \int_0^{2\pi} \left(\frac{1}{4} - \frac{1}{5} \right) d\varphi = \int_0^{2\pi} \frac{1}{20} d\varphi = \frac{1}{20} \cdot 2\pi = \frac{\pi}{10} \end{aligned}$$

Final Answer:

$$\text{Flux} = \frac{\pi}{10}$$

Question 5. Show that $\vec{F} = (8xe^{2z} + 5y^2)\vec{i} + (10xy - 3y^2e^{2z})\vec{j} + e^{2z}(8x^2 - 2y^3)\vec{k}$ is a conservative vector field. Find a function f such that $\vec{F} = \nabla f$.

[Answer]

We have:

$$\begin{aligned} \text{rot } \vec{F} &= \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) \vec{i} + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \right) \vec{j} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \vec{k} \\ &= \left[\frac{\partial}{\partial y} (e^{2z}(8x^2 - 2y^3)) - \frac{\partial}{\partial z} (10xy - 3y^2e^{2z}) \right] \vec{i} \\ &\quad + \left[\frac{\partial}{\partial z} (8xe^{2z} + 5y^2) - \frac{\partial}{\partial x} (e^{2z}(8x^2 - 2y^3)) \right] \vec{j} \end{aligned}$$

$$+ \left[\frac{\partial}{\partial x} (10xy - 3y^2 e^{2z}) - \frac{\partial}{\partial y} (8xe^{2z} + 5y^2) \right] \vec{k}$$

$$= [-6y^2 e^{2z} + 6y^2 e^{2z}] \vec{i} + [16xe^{2z} - 16xe^{2z}] \vec{j} + [10y - 10y] \vec{k} = \vec{0}.$$

Therefore, $\vec{F}$ is a conservative vector field. Choose $(x_0, y_0, z_0) = (0; 0; 0)$, the function f is defined by:

$$\begin{aligned} f(x, y, z) &= \int_0^x P(t, 0, 0) dt + \int_0^y Q(x, t, 0) dt + \int_0^z R(x, y, t) dt + C \\ &= \int_0^x (8te^0 + 0) dt + \int_0^y (10xt - 3t^2 e^0) dt + \int_0^z e^{2t} (8x^2 - 2y^3) dt + C \\ &= \int_0^x 8t dt + \int_0^y (10xt - 3t^2) dt + (8x^2 - 2y^3) \int_0^z e^{2t} dt + C \\ &= 4x^2 + (5xy^2 - y^3) + (8x^2 - 2y^3) \left(\frac{e^{2z} - 1}{2} \right) + C \\ &= 4x^2 + 5xy^2 - y^3 + \frac{(8x^2 - 2y^3)(e^{2z} - 1)}{2} + C \end{aligned}$$

Question 6. Evaluate $\iiint_V \sqrt{4x - x^2 - y^2 - z^2} dx dy dz$, where V is defined by $x^2 + y^2 + z^2 \leq 4x$.

[Answer]

+) Change of variables $\begin{cases} x = r \cos \alpha \sin \theta \\ y = r \sin \alpha \sin \theta \\ z = r \cos \theta \end{cases}$ with $\begin{cases} 0 \leq r \leq 2 \\ 0 \leq \theta \leq \pi \\ 0 \leq \alpha \leq 2\pi \end{cases}$

+) Replace in:

$$\begin{aligned}
 \iiint_V \sqrt{4x - x^2 - y^2 - z^2} \, dx \, dy \, dz &= \int_0^{2\pi} \int_0^\pi \int_0^2 \sqrt{4r - r^2 \sin^2 \theta} \cdot r^2 \sin \theta \, dr \, d\theta \, d\alpha \\
 &= \int_0^{2\pi} d\alpha \cdot \int_0^\pi \sin \theta \, d\theta \cdot \int_0^2 r^2 \sqrt{4 - r^2} \, dr \\
 &= 4\pi \int_0^2 r^2 \sqrt{4 - r^2} \, dr \\
 (\text{Đặt } r = 2 \sin \varphi \Rightarrow dr &= 2 \cos \varphi \, d\varphi) \\
 &= 4\pi \int_0^{\pi/2} (2 \sin \varphi)^2 \cdot \sqrt{4 - 4 \sin^2 \varphi} \cdot 2 \cos \varphi \, d\varphi \\
 &= 4\pi \int_0^{\pi/2} 16 \sin^2 \varphi \cos^2 \varphi \, d\varphi \\
 &= 64\pi \int_0^{\pi/2} \sin^2 \varphi \cos^2 \varphi \, d\varphi \\
 &= 16\pi \int_0^{\pi/2} \sin^2 2\varphi \, d\varphi \\
 &= 8\pi \int_0^{\pi/2} (1 - \cos 4\varphi) \, d\varphi \\
 &= 8\pi \left[\varphi - \frac{\sin 4\varphi}{4} \right]_0^{\pi/2} \\
 &= 8\pi \cdot \frac{\pi}{2} = 4\pi^2
 \end{aligned}$$

+) Conclusion

$$\boxed{4\pi^2}$$

Question 7. Evaluate the following integrals:

a) $\int_0^{+\infty} \frac{x^2 \sqrt{x}}{(x^3 + 1)^2} \, dx$

b) $\int_0^{+\infty} \frac{e^{-x} - 3^{-x}}{x} \, dx$

[Answer]

(a) $\int_0^{+\infty} \frac{x^2 \sqrt{x}}{(x^3 + 1)^2} \, dx$

Let $x = u^{1/3} \Rightarrow x^{5/2} = u^{5/6}$, $dx = \frac{1}{3}u^{-2/3}du$, and $x^3 + 1 = u + 1$ then:

$$\begin{aligned}
 \int_0^\infty \frac{x^2 \sqrt{x}}{(x^3 + 1)^2} \, dx &= \int_0^\infty \frac{x^{5/2}}{(x^3 + 1)^2} \, dx \\
 &= \int_0^\infty \frac{u^{5/6}}{(u + 1)^2} \cdot \frac{1}{3}u^{-2/3} \, du = \frac{1}{3} \int_0^\infty \frac{u^{1/6}}{(u + 1)^2} \, du
 \end{aligned}$$

Evaluate the integral using the beta function identity:

$$\int_0^\infty \frac{u^{p-1}}{(1 + u)^{p+q}} \, du = B(p, q) = \frac{\Gamma(p)\Gamma(q)}{\Gamma(p + q)}$$

So:

$$\begin{aligned} \frac{1}{3} \int_0^\infty \frac{u^{1/6}}{(u+1)^2} du &= \frac{1}{3} B\left(\frac{7}{6}, \frac{5}{6}\right) = \frac{1}{3} \cdot \frac{\Gamma\left(\frac{7}{6}\right) \Gamma\left(\frac{5}{6}\right)}{\Gamma(2)} \\ \Gamma\left(\frac{1}{6}\right) \Gamma\left(\frac{5}{6}\right) &= \frac{\pi}{\sin\left(\frac{\pi}{6}\right)} = \frac{\pi}{1/2} = 2\pi \Rightarrow \Gamma\left(\frac{7}{6}\right) = \frac{1}{6} \Gamma\left(\frac{1}{6}\right) \\ \Rightarrow \text{Answer} &= \frac{1}{3} \cdot \frac{\Gamma\left(\frac{7}{6}\right) \Gamma\left(\frac{5}{6}\right)}{1} = \frac{\pi}{3 \sin\left(\frac{5\pi}{6}\right)} = \frac{\pi}{3 \cdot \frac{1}{2}} = \frac{2\pi}{3} \\ \text{(b)} \quad &\int_0^{+\infty} \frac{e^{-x} - 3^{-x}}{x} dx \end{aligned}$$

+) We have:

$$\frac{e^{-x} - 3^{-x}}{x} = \frac{e^{-x} - e^{-x \ln 3}}{x} = \frac{e^{-yx}}{x} \Big|_{y=\ln 3}^{y=1} = \int_{\ln 3}^1 \left(\frac{e^{-yx}}{x} \right)'_y dy = \int_1^{\ln 3} e^{-yx} dy$$

Then:

$$I = \int_0^{+\infty} \frac{e^{-x} - 3^{-x}}{x} dx = \int_0^{+\infty} \left(\int_1^{\ln 3} e^{-yx} dy \right) dx$$

+) Consider the improper integral depending on the parameter $\int_0^{+\infty} e^{-yx} dx$ với $y \in [1; \ln 3]$

- $f(x, y) = e^{-yx}$ continuous on $[0; +\infty) \times [1; \ln 3]$.

- $|e^{-yx}| \leq e^{-x}$, but $\int_0^{+\infty} e^{-x} dx$ converges.

$\Rightarrow \int_0^{+\infty} e^{-yx} dx$ uniformly converges according to Weierstrass.

+) Therefore, we have:

$$I = \int_1^{\ln 3} \int_0^{+\infty} e^{-yx} dx dy = \int_1^{\ln 3} \left[\frac{-e^{-yx}}{y} \right]_0^{+\infty} dy = - \int_1^{\ln 3} \frac{1}{y} dy = \ln |y| \Big|_1^{\ln 3} = \ln \ln 3.$$

Question 8. Evaluate $\int_C (6x\sqrt{x^2+1} - 2xy^3) dx + (y^3 - 3x^2y^2) dy$, where C is the curve $y = \sqrt[4]{1-x^3}$ from $A(1; 0)$ to $B(0; 1)$.

[Answer]

+) Let

• L be the line from $A(1, 0)$ to $B(0, 1)$.

• $P = 6x\sqrt{x^2+1} - 2xy^3$, $Q = y^3 - 3x^2y^2$.

$$\Rightarrow \int_C P dx + Q dy = \int_{C+L} P dx + Q dy - \int_L P dx + Q dy$$

+) Since $C + L$ is closed:

$$\Rightarrow \int_{C+L} P dx + Q dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy$$

where region $D: \begin{cases} 0 \leq x \leq 1, \\ 0 \leq y \leq \sqrt[4]{1-x^3}. \end{cases}$

$$\frac{\partial Q}{\partial x} = -6xy^2, \quad \frac{\partial P}{\partial y} = -6xy^2 \Rightarrow \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 0.$$

$$\Rightarrow \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy = \iint_D 0 dx dy = 0$$

+) Compute $\int_L P dx + Q dy$:

$$L: x = t, \quad y = 1 - t, \quad t: 1 \rightarrow 0, \quad dx = dt, \quad dy = -dt$$

$$\begin{aligned} & \int_L (6x\sqrt{x^2+1} - 2xy^3) dx + (y^3 - 3x^2y^2) dy \\ &= \int_1^0 [6t\sqrt{t^2+1} - 2t(1-t)^3] dt + [(1-t)^3 - 3t^2(1-t)^2] (-dt) \\ &= \int_1^0 [6t\sqrt{t^2+1} - 2t(1-3t+3t^2-t^3) - (1-3t+3t^2-t^3) + 3t^2(1-2t+t^2)] dt \\ &= \int_1^0 [6t\sqrt{t^2+1} + t + 6t^2 - 11t^3 + 5t^4 - 1] dt \\ &= \int_0^1 [-6t\sqrt{t^2+1} - t - 6t^2 + 11t^3 - 5t^4 + 1] dt \\ &= \int_0^1 [-6t\sqrt{t^2+1} - 5t^4 + 11t^3 - 6t^2 - t + 1] dt \\ &= \left[-2(t^2+1)^{3/2} - \frac{5t^5}{5} + \frac{11t^4}{4} - \frac{6t^3}{3} - \frac{t^2}{2} + t \right]_0^1 \\ &= -4\sqrt{2} + \left(-1 + \frac{11}{4} - 2 - \frac{1}{2} + 1 + 2 \right) \\ &= -4\sqrt{2} + \left(\frac{-4 + 11 - 8 - 2 + 4 + 8}{4} \right) \\ &= -4\sqrt{2} + \frac{9}{4} \end{aligned}$$

+) Combine:

$$\begin{aligned} & \int_C (6x\sqrt{x^2+1} - 2xy^3) dx + (y^3 - 3x^2y^2) dy \\ &= 0 - \left(-4\sqrt{2} + \frac{9}{4} \right) \\ &= 4\sqrt{2} - \frac{9}{4} \end{aligned}$$

+) Conclusion

$$\boxed{4\sqrt{2} - \frac{9}{4}}$$

Question 9. Find the area of the region enclosed by:

$$9x^2 - 4xy + y^2 = 4$$

[Answer]

+) We have: $9x^2 - 4xy + y^2 = 4 \Leftrightarrow 5x^2 + (2x - y)^2 = 4$

$$\text{Let: } \begin{cases} x = \frac{r \cdot \cos(\phi)}{\sqrt{5}} \\ 2x - y = r \cdot \sin(\phi) \end{cases} \Leftrightarrow \begin{cases} x = \frac{r \cdot \cos(\phi)}{\sqrt{5}} \\ y = \frac{2r \cdot \cos(\phi)}{\sqrt{5}} - r \cdot \sin(\phi) \end{cases}$$

Now, we have region enclosed by: $D \begin{cases} 0 \leq r \leq 2 \\ 0 \leq \phi \leq 2\pi \end{cases}$

$$\text{We have } |J| = \begin{vmatrix} x'_r & x'_\phi \\ y'_r & y'_\phi \end{vmatrix} = \begin{vmatrix} \frac{\cos(\phi)}{\sqrt{5}} & \frac{-r \cdot \sin(\phi)}{\sqrt{5}} \\ \frac{2 \cdot \cos \phi}{\sqrt{5}} - \sin \phi & \frac{-2r \cdot \sin(\phi)}{\sqrt{5}} - r \cdot \cos(\phi) \end{vmatrix} = \frac{r}{\sqrt{5}}$$

+) The area of this region is:

$$S = \iint_D \frac{r}{\sqrt{5}} dr d\phi = \int_0^{2\pi} d\phi \int_0^2 \frac{r}{\sqrt{5}} dr = \frac{2\pi}{\sqrt{5}} \cdot \left. \frac{r^2}{2} \right|_0^2 = \frac{4\pi\sqrt{5}}{5}$$

+) Conclusion: $\frac{4\pi\sqrt{5}}{5}$